Homework	7
SID	301
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Attempted	0 out of 12

Q#	Ref/Section	Exer #	Resources
Q1	Rosen/Section 4.3	6	

How many zeros are there at the end of 100!?

Answer:
Trailing zeros come from factors of $10 = 2 \cdot 5$.
There are more factors of 2 than 5 in $100!$, we will count the number of 5s:

$$\lfloor\frac{100}{5}\rfloor + \lfloor\frac{100}{25}\rfloor + \lfloor\frac{100}{125}\rfloor = 20 + 4 + 0 = 24$$

Q#	Ref/Section	Exer #	Resources
Q2	Rosen/Section 4.3	14	

Which positive integers less than 12 are relatively prime to 12?

Answer:

We have the positive integers are less than 12:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

We will check which gcd 1 with 12:

- gcd $(1,12) = 1$
- gcd $(5,12) = 1$
- gcd $(7,12) = 1$
- gcd $(11,12) = 1$

Finaly: the answer is: 1, 5, 7, 11

Q#	Ref/Section	Exer #	Resources
Q3	Rosen/Section 4.3	30	

If the product of two integers is $2^7 3^8 5^2 7^{11}$ and their greatest common divisor is $2^3 3^4 5$, what is their least common multiple?

Answer:

Step 1: We will use: $ab = gcd(a, b) \cdot lcm(a, b)$
Step 2: $lcm(a, b) = \frac{ab}{\gcd(a,b)}$
Step 3: $lcm(a, b) = \frac{2^7 3^8 5^2 7^{11}}{2^3 3^4 5} = 2^{7-3} 3^{8-4} 5^{2-1} 7^{11}$
Final Step: $lcm(a, b) = 2^4 3^4 5 7^{11}$

Q#	Ref/Section	Exer #	Resources
Q4	Rosen/Section 4.3	42	

Use the extended Euclidean algorithm to express gcd*(*252, 356*)* as a linear combination of 252 and 356.

Answer:

We will find the gcd which is step 1:

$$252 = 2(104) + 44$$
$$104 = 2(44) + 16$$
$$44 = 2(16) + 12$$
$$16 = 1(12) + 4$$
$$12 = 3(4) + 0$$

Step 2: $\gcd(252,356) = 4$
Step 3: we will work backward→ $4 = 16 - 12$. Since: $12 = 44 - 2(16)$.
Step 4: we will get $4 = 16 - (44 - 2 \cdot 16) = 3 \cdot 16 - 44$.

Since

$$16 = 104 - 2(44)$$
$$4 = 3(104 - 2 \cdot 44) - 44 = 3 \cdot 104 - 7 \cdot 44$$

Since

$$44 = 252 - 2(104)$$
$$4 = 3 \cdot 104 - 7(252 - 2 \cdot 104) = 17 \cdot 104 - 7 \cdot 252$$

Since

$$104 = 356 - 252$$
$$4 = 17(356 - 252) - 7 \cdot 252$$
$$4 = 17 \cdot 356 - 24 \cdot 252$$

Finaly: $4 = -24(252) + 17(356)$

Final Answer: $\gcd(252, 356) = 4 = -24 \cdot 252 + 17 \cdot 356$

#	Ref/Section	Exer #	Resources
Q5	Rosen/Section 4.3	46(c,d)	

Find the smallest positive integer with exactly n different positive factors when n is

c) 5
Answer:

If a number has prime factorization: $p_1^{a1} p_2^{a2} \cdots p_k^{ak}$, then the number of positive divisors is$(a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$. To get exactly 5 divisors, we need: $5 = 5$. So the number must be of the form p^4. The smallest such number is: $2^4 = 16$

d) 6
Answer:

We need: $6 = 6$or$6 = 3 \cdot 2$. So the possibilities are 1- p^5 2-$p^2q$with distinct primes p, q After that: we check the smallest:

- $2^5 = 32$
- $2^2 \cdot 3 = 12$

Finaly: The smaller is 12.

Q#	Ref/Section	Exer #	Resources
Q6	Rosen/Section 4.4	6(b,c)	

Find an inverse of a modulo m for each of these pairs of relatively prime integers using the method followed in Example 2.

b) $a = 34$, $m = 89$

Answer: we will use the Euclidean algorithm:

$$34 = 1(21) + 13$$
$$21 = 1(13) + 8$$
$$13 = 1(8) + 5$$
$$8 = 1(5) + 3$$
$$5 = 1(3) + 2$$
$$3 = 1(2) + 1$$

Now we will back substitute:

$$3 - (5 - 3) = 2 \cdot 3 - 5$$
$$2(8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5$$
$$2 \cdot 8 - 3(13 - 8) = 5 \cdot 8 - 3 \cdot 13$$
$$5(21 - 13) - 3 \cdot 13 = 5 \cdot 21 - 8 \cdot 13$$
$$5 \cdot 21 - 8(34 - 21) = 13 \cdot 21 - 8 \cdot 34$$
$$13(89 - 2 \cdot 34) - 8 \cdot 34 = 13 \cdot 89 - 34 \cdot 34$$

Then we will get: $-34 \cdot 34 \equiv 1(mod 89)$.
the inverse of 34 modulo 89 is: $-34 \equiv 55(\text{mod}89)$

Final Answer: $34^{-1} \equiv 55\ (mod\ 89)$

c) $a = 144$, $m = 233$

Answer: we will use extended Euclidean algorithm that we obtained:
$1 = 89 \cdot 144 - 55 \cdot 233$. So $89 \cdot 144 \equiv 1(mod 233)$

Finaly: $144^{-1} \equiv 89\ (mod 233)$

Q#	Ref/Section	Exer #	Resources
Q7	Rosen/Section 4.4	12(a,b)	

Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 6.

a) $34x \equiv 77 \pmod{89}$

Answer: From previous question we will multiply both sides by 55:
$x \equiv 55 \cdot 77 (mod 89)$
→ $55 \cdot 77 = 4235$
Now we will reduce mod 89: $4235 - 89 \cdot 47 = 4235 - 4183 = 52$

Final Answer: $x \equiv 52\ (mod\ 89)$

b) $144x \equiv 4 \pmod{233}$

Answer: Here we will multiply both sides by 89:
$x \equiv 89 \cdot 4 (mod 233)$
→ $89 \cdot 4 = 356$
We will reduce mod 233: $356 - 233 = 123$

Final Answer: $x \equiv 123\ (mod\ 233)$

Q#	Ref/Section	Exer #	Resources
Q8	Rosen/Section 4.4	34	

Use Fermat's little theorem to find 23^{1002} **mod** 41.

Answer: $23^{40} \equiv 1\ (mod 41)$

Now we will reduce the exponent: $1002 = 40 \cdot 25 + 2$

So that:

$$23^{1002} = 23^{40 \cdot 25 + 2} = (23^{40})^{25} \cdot 23^2$$
$$\equiv 1^{25} \cdot 23^2 (mod 41)$$
$$23^2 = 529$$

We will reduce 529 mod 41: $529 - 41 \cdot 12 = 529 - 492 = 37$

Finaly: $23^{1002} \equiv 37\ (mod\ 41)$

Q#	Ref/Section	Exer #	Resources
Q9	Rosen/Section 5.1	4	

Let $P(n)$ be the statement that $1^3 + 2^3 + \cdots + n^3 = (n(n+1)/2)^2$ for the positive integer n.

a) What is statement $P(1)$?

Answer: $1^3 = (\frac{1(1+1)}{2})^2 \rightarrow 1 = (\frac{2}{2})^2 = 1$

Finaly: $1^3 = (\frac{1 \cdot 2}{2})^2$

b) Show that $P(1)$ is true, completing the basis step of the proof.

Answer: $1^3 = 1$ and $(\frac{1(2)}{2})^2 = 1^2 = 1$. So both sides are equal

Final Answer: True

c) What is the inductive hypothesis?

Answer: We will assume $P(k)$ is true for some positive integer k. That is:

$$1^3 + 2^3 + \cdots + k^3 = (\frac{k(k+1)}{2})^2$$

This is the inductive hypothesis.

d) What do you need to prove in the inductive step?

Answer: we will approve that $P(k+1)$is true:

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3 = (\frac{(k+1)(k+2)}{2})^2$$

e) Complete the inductive step, identifying where you use the inductive hypothesis.

Answer: we will Start with the left-hand side of $P(k+1)$:

$$1^3 + 2^3 + \cdots + k^3 + (k+1)^3$$

We will use the inductive hypothesis,

$$= (k+1)^2 (\frac{k^2}{4} + k + 1)$$

$$= (k+1)^2 (\frac{k^2 + 4k + 4}{4})$$

$$= (k+1)^2 (\frac{(k+2)^2}{4})$$

$$= (\frac{(k+1)(k+2)}{2})^2$$

This is exactly the right-hand side of $P(k+1)$.

Therefore $P(k+1)$is true.

f) Explain why these steps show that this formula is true whenever n is a positive integer.

Answer:

We proved:

1. Basis step: $P(1)$is true.
2. Inductive step: If $P(k)$is true, then $P(k+1)$is true.

Therefore, by mathematical induction, $P(n)$is true for every positive integer n.

Q#	Ref/Section	Exer #	Resources
Q10	Rosen/Section 5.1	28	

Prove that n^2 - $7n$ + 12 is nonnegative whenever n is an integer with $n \geq 3$

Answer: We use induction on n.

Basis step: $n = 3 \rightarrow 3^2 - 7(3) + 12 = 9 - 21 + 12 = 0$
So it is nonnegative.

Inductive hypothesis

Assume for some $k \geq 3$, $\rightarrow k^2 - 7k + 12 \geq 0$

Inductive step

We must show $\rightarrow (k+1)^2 - 7(k+1) + 12 \geq 0$

We compute: $(k+1)^2 - 7(k+1) + 12 = k^2 + 2k + 1 - 7k - 7 + 12$
$= k^2 - 5k + 6$

Now rewrite this as: $k^2 - 5k + 6 = (k^2 - 7k + 12) + (2k - 6)$

By the inductive hypothesis, $k^2 - 7k + 12 \geq 0$

and since $k \geq 3$, $\rightarrow 2k - 6 \geq 0$

So, $(k+1)^2 - 7(k+1) + 12 \geq 0$

Thus the statement is true for $k + 1$.

Hence, by induction,

$n^2 - 7n + 12 \geq 0 \; for \; all \; integers \; n \geq 3$

Q#	Ref/Section	Exer #	Resources
Q11	Rosen/Section 5.1	44	

Prove that if $A_1, A_2, \ldots, A_n$ and B are sets, then
$(A_1 - B) \cup (A_2 - B) \cup \cdots \cup (A_n - B) = (A_1 \cup A_2 \cup \cdots \cup A_n) - B$.

Answer:
Let x be any element: $x \in (A1 - B) \cup (A2 - B) \cup \cdots \cup (An - B)$

Iff $x \in A_i - B \; for \; some \; i \in \{1,2,\ldots,n\}$
Iff $x \in Ai \; and \; x \notin B for \; some \; i$
Iff $x \in A1 \cup A2 \cup \cdots \cup An \; and \; x \notin B$
Iff $x \in (A1 \cup A2 \cup \cdots \cup An) - B$

Final Answer: $(A1 - B) \cup (A2 - B) \cup \cdots \cup (An - B) = (A1 \cup A2 \cup \cdots \cup An) - B$

Q#	Ref/Section	Exer #	Resources
Q12	Rosen/Section 5.1	70	

Use mathematical induction to prove that $G(n) \leq 2n - 4$ for $n \geq 4$.
[*Hint:* In the inductive step, have a new person call a particular person at the start and at the end.]

Answer:

We prove by induction that $G(n) \leq 2n - 4$for $n \geq 4$.

Base case (n = 4):
It is possible for 4 people to share all gossip in 4 calls, so $G(4) \leq 4 = 2(4) - 4$

Inductive hypothesis:
Assume $G(k) \leq 2k - 4$for some $k \geq 4$.

Inductive step:
Consider $k + 1$people. Let the new person call someone at the start (1 call), then the original $k$people exchange gossip in at most $2k - 4$calls, and finally the new person calls again at the end (1 call).

So, $G(k + 1) \leq 1 + (2k - 4) + 1 = 2k - 2 = 2(k + 1) - 4$

By induction, $G(n) \leq 2n - 4 for\ all\ n \geq 4$.